Jia, Ye

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EDUCATION

CHONGQING UNIVERSITY OF ARTS AND SCIENCES Chongqing, China
Bachelor of Engineering 09/2018 – 06/2022

- Major: Robot Engineering
- Average Score: 3.05/4
- 3rd Prize in the 4th Robot and Intelligent Device Application Design Competition of Chongqing University of Arts and Sciences
- 2nd Prize in the 3d Digital Innovation Design Competition of Chongqing University of Arts and Sciences
- Gold Medal Award in the 5th China "Internet +" College Students Innovation and Entrepreneurship Competition, Chongqing University of Arts and Sciences, main track
- Bronze Medal Award in the 5th China "Internet +" College Students Innovation and Entrepreneurship Competition Chongqing Area trial

PUBLICATIONS

Design of Intelligent Multi-layer Three-dimensional Bicycle Parking Garage

- Wang, C., **Jia, Y**., Zhao, L., & Guo, P. (2020, October). Design of Intelligent Multi-layer Three-dimensional Bicycle Parking Garage. In Journal of Physics: Conference Series (Vol. 1624, No. 6, p. 062019). IOP Publishing.

Design Based on STM32 Anti-compression Airbag Helmet

- Hu Xuerui, **Ye Jia**. Textile Reports ,2020(3):97-98. DOI:10.3969/j.issn.1005-6289.2020.03.039.

Rolling Bearing Fault Diagnosis Using ResNet Family Deep-Learning Models.

- Wu, G.; Ji, X.; Yang, G.; **Jia, Y**.; Cao, C. Processes 2023, 11, 1527. https://doi.org/10.3390/pr11051527

SIGNIFICANT SCIENTIFIC PROJECTS

VIRTUAL SIMULATION OF FRUIT PICKING ROBOT BASED ON UNITY3D 08/2020 - 11/2020
Research Assistant

- **Introductions:** This project is a sub-project designed for picking robot, mainly through the form of software modeling simulation to explore the possibility of fruit picking.
- **Professional Duties:** 1. Took charge of searching for relevant design literatures of picking robots at home and abroad; 2. Wrote the paper

DESIGN OF INTELLIGENT MULTI-LAYER THREE-DIMENSIONAL BICYCLE 01/2020 - 07/2020
Core Participant

- **Background Introductions:** The project aims to design an intelligent bicycle stereo parking lot, given the increasingly scarce urban land resources. This design mainly applies information technology and SCM technology. The project is divided into two parts: one is the model building and mechanical simulation of the garage; the other is the control and management program of the bicycle garage based on Arduino. Finally, the feasibility of the scheme is verified.
- **Professional Duties:** Took charge of writing the control and management program with Arduino and realized the intelligent management and control of the bicycle garage through virtual simulation.
- **Achievement:** The manuscript of this project has been published on Journal of Physics: Conference Series

DESIGN OF ANTI-EXTRUSION AIRBAG HELMET BASED ON STM32 11/2019 - 03/2020
Core Participant

- **Background Introductions:** Based on the analysis and selection of the application site of the product, the project selects the single-chip microcomputer. It then installs the protective module based on the helmet appearance design, being both protective and practical.
- **Professional Duties:** 1. Took charge of designing the airbag protection module in the helmet, designed the ejection logic and trigger circuit of the airbag; 2. Demonstrated the rationality and necessity of the selection of each component; 3. Made the whole helmet airbag protection system more reliable through the programming of STM32 MCU
- **Achievement**: The manuscript of this project has been published on Textile Reports

IDENTIFICATION OF GREEN CITRUS BASED ON YOLO- V2 03/2021
Core Participant

- **Background Introduction:** In this project, I realized the recognition process of the deep learning model through the operation learning of Matlab and in-depth learning of YOLO- V2 and trained the recognition network with green citrus as the learning target.
- **Professional Duties: 1.** Collected tangerines related pictures, and used MATLAB to box selection of tangerines in the collected pictures; 2. Set parameters of deep learning model and train neural network
- Achievements:Completed the recognition of static citrus images, and the recognition matching degree reached 60%-90%%

DEEP LEARNING AND NEURAL NETWORK MODELS ASSOCIATED WITH YOLO 04/2021 - 07/2021
Core Participant

- **Background Introduction:**This project mainly studies the identification of pear flower and green orange based on YoloV4 under Darknet framework and PyTorch framework, so as to better identify different objects and completed the grasping action of the mechanical arm.
- **Professional Duties:** 1. Modified the original code; 2. Collected and marked pictures
- **Achievements:**
- Implemented detection of YOLOv4 network in Darknet framework and PyTorch framework;
- Implemented YOLOv3 and YOLOv5 algorithms for citrus and pear detection;
- Made changes to the yolov4-PyTorch version of the code.

USE OF ROS SYSTEM IN UBANTU ENVIRONMENT 08/2021 - 12/2021

Core Participant

- **Background Introduction:** The main purpose of this study is to realize the control communication between machine vision recognition and manipulator grasping. In this study, I also learned the use of ROS system in the UBANTU environment.
- **Professional Duties:** 1. Configured the ROS system installation environment and completed the ROS installation; 2. Learnt about ROS system related knowledge; 3. Completed the communication between ROS system and depth camera, industrial camera and robot
- **Achievements:**
- Configured the Ubuntu OS using an independent VM and installed ros_20.04 and 18.04 versions in this environment;
- Realized TurtleSim running under ROS software;
- Mastered the knowledge of the ROS system.

ROLLING BEARING FAULT DIAGNOSIS BASED ON DEEP LEARNING NURYUAL NETWORK

Core Participant 03/2022 - 07/2022

- **Background Introduction:** The main purpose of this research is to apply a deep learning model for rolling bearing fault diagnosis. In this research, I proposed an improved Residual Network50(Resnet50) with Squeeze Extraction(SE) methods and applied it to detect broken rooling bearings using the original date of Case western reserve university(CWRU) to train the model.
- **Professional Duties:** 1. I transformed the one-dimensional signal data of the two-dimensional RGB image by using continuous wavelet transform to enlarge fault features from the one-dimensional signal data; 2. I improved the Resnet50 model and combine it with Squeeze Extraction(SE) blocks; 3. I trained the improved model using data from CWRU. 4. Main auther of the paper.
- **Achievements:** Wu, G.; Ji, X.; Yang, G.; **Jia, Y**.; Cao, C. Signal-to-Image: Rolling Bearing Fault Diagnosis Using ResNet Family Deep-Learning Models. *Processes* **2023**, *11*, 1527. https://doi.org/10.3390/pr11051527
- Improved and trained the improved Resnet50 model.
- Proposed and realized the data handle method using the continuous wavelet transform.

K-CUBE VR

Research Assistant 08/2022-- present

- **Background Introduction:** The key idea of K-Cube VR is to combine the immersion of virtual reality and the structure of a knowledge graph to present educational knowledge in such a way that facilitates association, exploration and engagement which benefits learning.
- **Professional Duties:** 1. Assist colleagues to improve K-Cube VR; 2. Conduct the research on the Enhance Point-and-Teleport Locomotion user study.
- **Achievements:**
  - ✧ Submitted the paper titled "Towards an Edu-Metaverse of Knowledge: Immersive Exploration of University Courses" manuscript successfully.(2nd author)
  - ✧ Polishing the paper titled "Enhance Point-and-Teleport Locomotion for Better Spatial Understanding and Memory in Virtual Reality", and plan to submit it on June. (1st author)

MAJOR EXTRACURRICULAR ACTIVITIES

STUDENT OFFICER, CULTURAL ACTIVITIES DEPARTMENT, COMMITTEE OF CHONGQING UNIVERSITY OF ARTS AND SCIENCES, COMMUNIST YOUTH LEAGUE 09/2018 - 06/2019

- Performed and organized daily performance activities
- Assisted in organizing medium and large cultural activities in the university;
- Wrote activity planning proposals, debriefing reports and activity reports;
- Took charge of the scheduling of stage sound consoles, personnel allocation and order maintenance;

SKILLS

- Proficient in office software, UG, Solid Works and other professional software；
- Proficient in C, C++, C#, Python and other development languages；
- Familiraised with the Unity3D development, developed the Enhanced teleportation technique test environment.